

Wazuh + Sysmon Threat Detection Report

Objective:

The objective of this lab was to integrate Microsoft Sysmon with the Wazuh SIEM to improve Windows endpoint visibility and detect attacker behavior using process creation and Windows event logs.

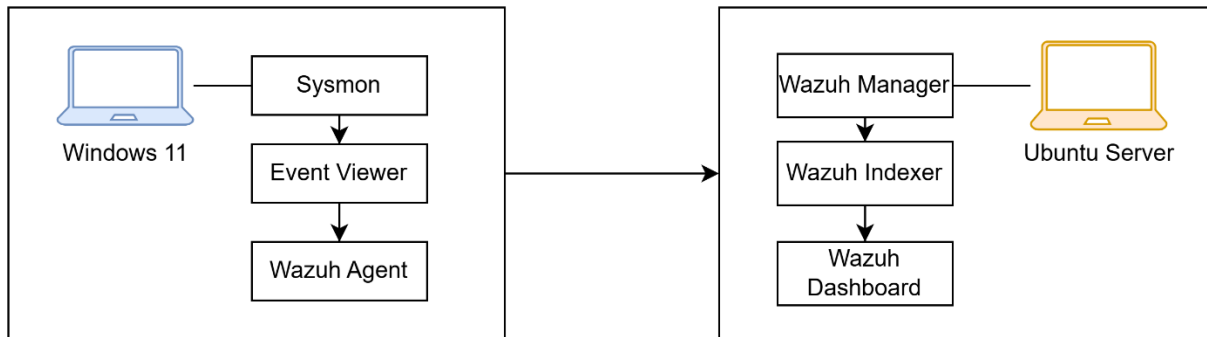
Lab Environment

Component	Details
SIEM	Wazuh 4.12
Manager OS	Ubuntu Server
Agent OS	Windows 11 Home
Log Source	Sysmon Event Logs
Network	Local Home Lab

Tools Used

- Wazuh Manager
- Wazuh Dashboard
- Wazuh Windows Agent
- Microsoft Sysmon
- SwiftOnSecurity Sysmon Configuration
- PowerShell
- Windows Event Viewer

Lab Architecture



Detection Testing

The following activities were performed

Test 1: Process Creation Event

Executed: cmd.exe

Expected: Sysmon Event ID 1

Observed: Process Creation Event received by Wazuh

Test 2: Powershell Execution

Executed: powershell.exe

Observed:

- PowerShell process detected
- Process metadata collected
- User information logged

Test 3: Network Utility

Executed: net user

Observed: Windows command execution successfully detected inside Wazuh Threat Hunting.

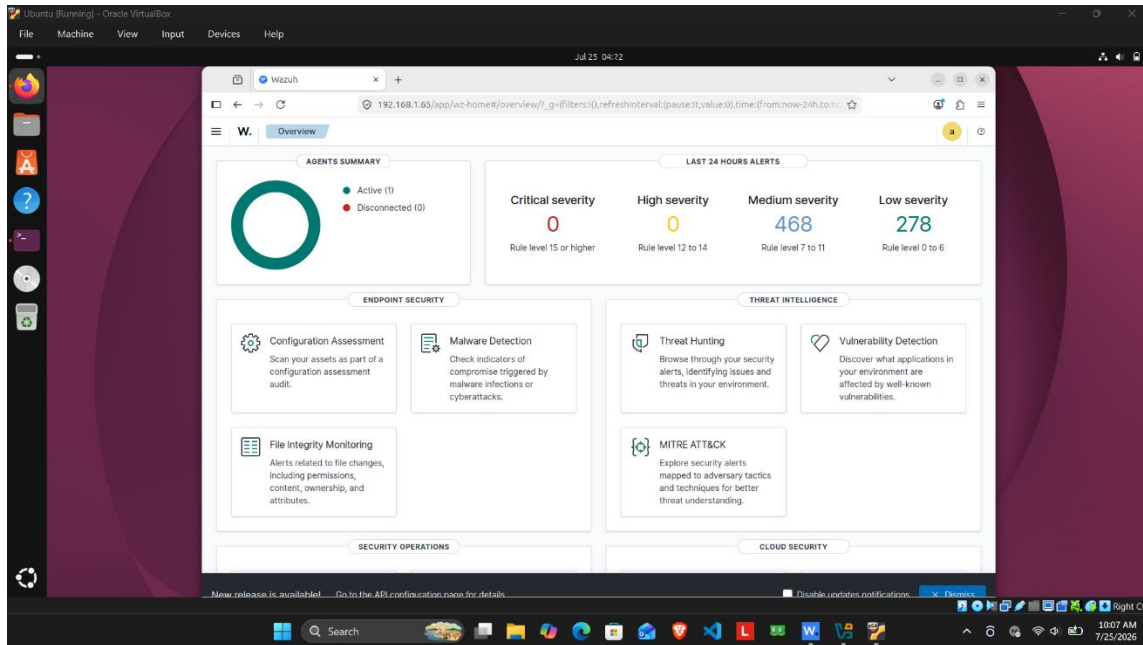
Test 4: Command Prompt

Executed: ipconfig

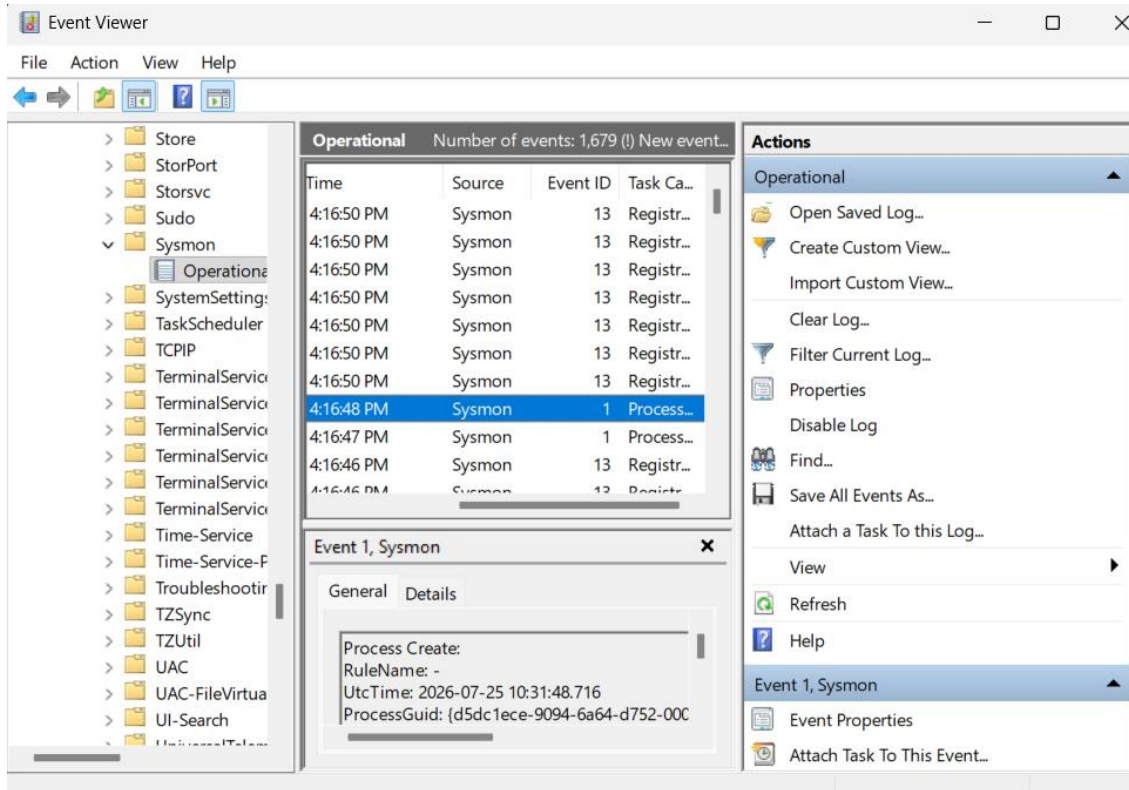
Observed: Process creation event generated.

Evidence:

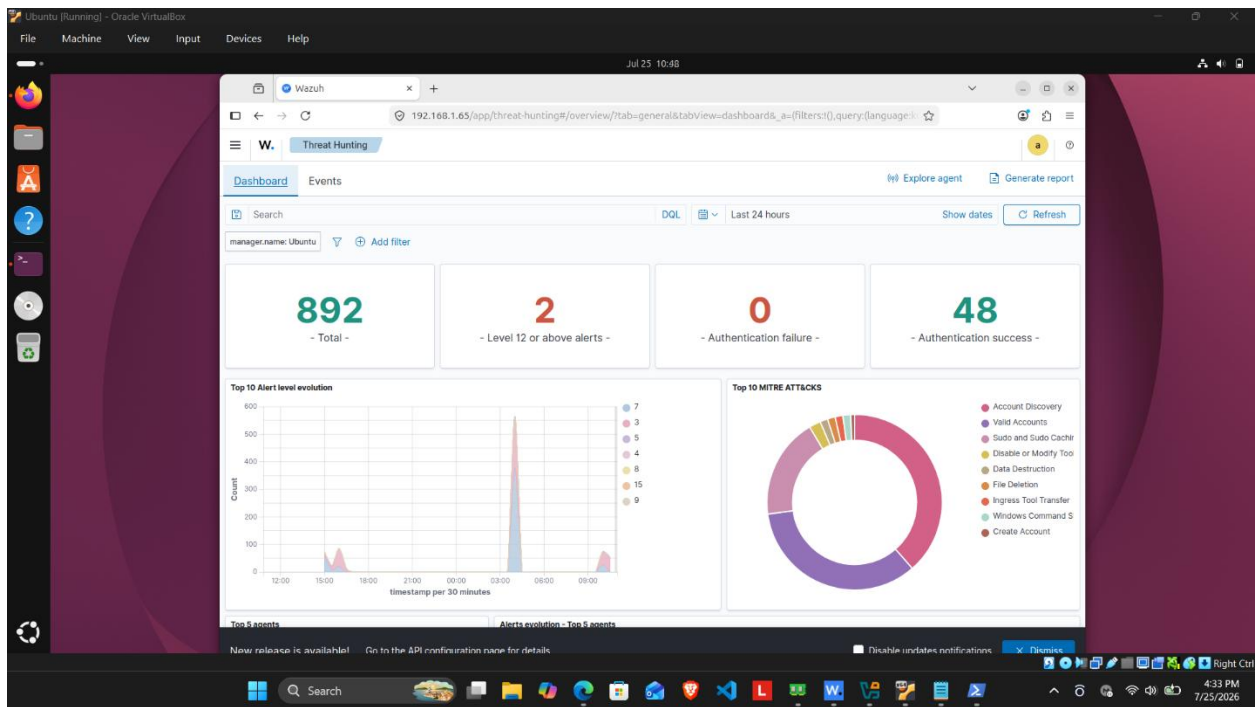
Wazuh Overview



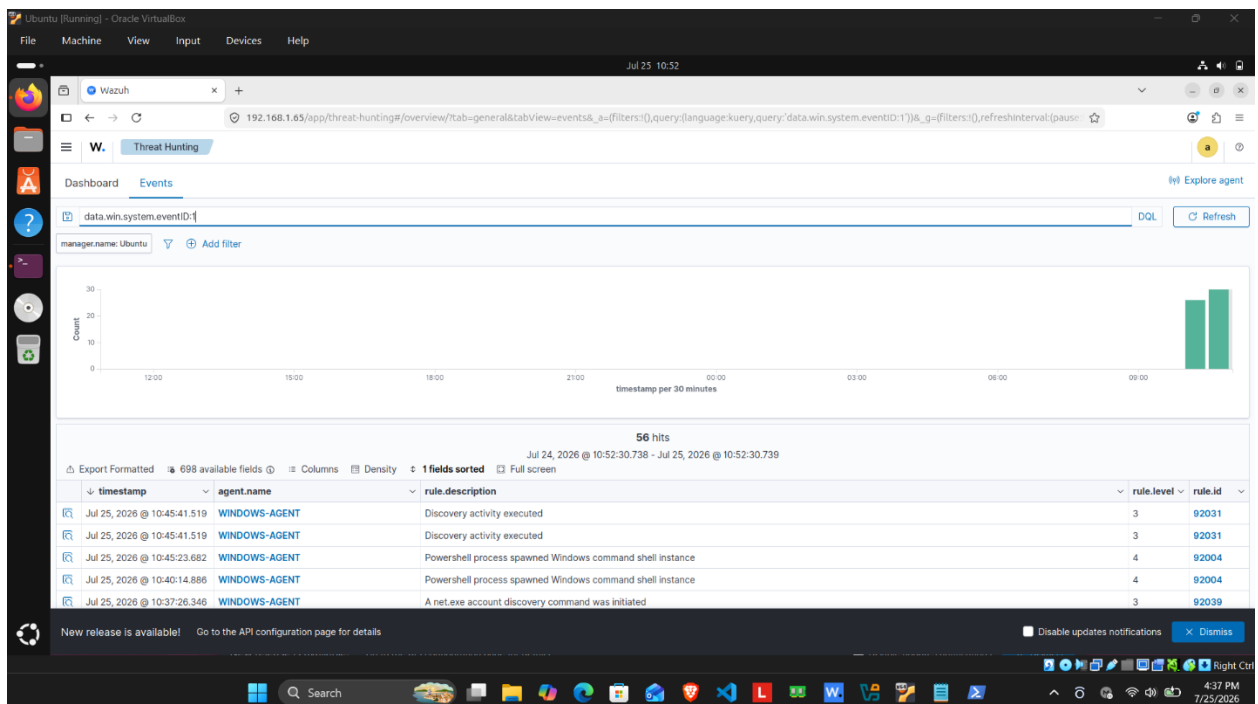
Sysmon Operational Log in Event Viewer



Threat Hunting Dashboard



data.win.system.eventID:1 search results.



Process details of cmd.exe

The screenshot shows the Wazuh dashboard interface. The main panel displays a list of events for the rule MITRE ATT&CK. The 'Document Details' panel on the right shows the following information:

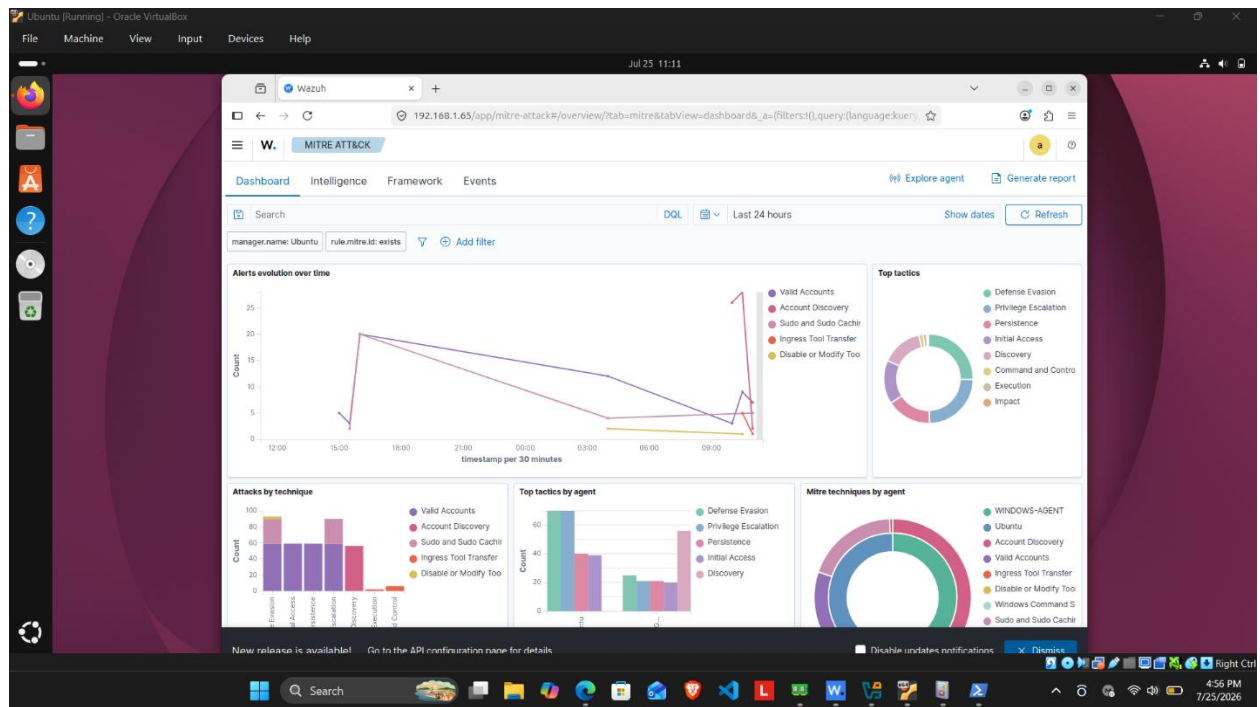
Field	Value
_index	wazuh-alerts-4.x-2026.07.25
agent.id	001
agent.ip	192.168.1.70
agent.name	WINDOWS-AGENT
data.win.eventdata.commandline	"C:\\Windows\\system32\\cmd.exe"
data.win.eventdata.company	Microsoft Corporation
data.win.eventdata.currentDirectory	C:\\Users\\khanal\\
data.win.eventdata.description	Windows Command Processor
data.win.eventdata.fileVersion	10.0.26100.8737 (WinBuild.160101.0000)
data.win.eventdata.hashes	MD5:1F4D183744499452622E8B04058B743E SHA256:65EC268A00972B85C6A4222985D47CAEAE44A34806C1466782914FD743D9_1MPHASH:0108165E4C37F48466103080708C9423
data.win.eventdata.image	C:\\Windows\\system32\\cmd.exe
data.win.eventdata.integrity	Medium
data.win.eventdata.logonGuid	{d5d1e3ce-359c-6a64-59f4-0f1b00000000}
data.win.eventdata.logonId	0x1b0ff459
data.win.eventdata.originalFilename	Cmd.Exe

Process details of net user

The screenshot shows the Wazuh dashboard interface. The main panel displays a list of events for the rule MITRE ATT&CK. The 'Document Details' panel on the right shows the following information:

Field	Value
_index	wazuh-alerts-4.x-2026.07.25
agent.id	001
agent.ip	192.168.1.70
agent.name	WINDOWS-AGENT
data.win.eventdata.commandline	C:\\WINDOWS\\system32\\net user guest
data.win.eventdata.company	Microsoft Corporation
data.win.eventdata.currentDirectory	C:\\Program Files (x86)\\ossec-agent\\
data.win.eventdata.description	Net Command
data.win.eventdata.fileVersion	10.0.26100.7019 (WinBuild.160101.0000)
data.win.eventdata.hashes	MD5:97EF48CB1E02DFE8B51672B7F8B1872 SHA256:68A0631C2F9ED0B8E70699AB043A5CE95851589569C5D15A0B01BA5725E131_1MPHASH:FEB41A7286568863C8C539552568F8
data.win.eventdata.image	C:\\Windows\\system32\\net1.exe
data.win.eventdata.integrity	System
data.win.eventdata.logonGuid	{d5d1e3ce-bf43-6a5d-e793-000000000000}
data.win.eventdata.logonId	0x3e7
data.win.eventdata.originalFilename	net1.exe

MITRE ATT&CK Dashboard



MITRE ATT&CK Events

The screenshot displays the MITRE ATT&CK Events view within the Wazuh interface. The view shows a table of events with the following columns:

- timestamp
- agent.name
- rule.mitre.id
- rule.mitre.tactic
- rule.description
- rule.level
- rule.id

The table contains 60 hits, sorted by timestamp. The events are filtered by the rule.mitre.id T1087 and T1059.003. The table shows a list of events with their respective timestamps, agent names, rule IDs, tactics, descriptions, levels, and rule IDs.

timestamp	agent.name	rule.mitre.id	rule.mitre.tactic	rule.description	rule.level	rule.id
Jul 25, 2026 @ 11:09:06.906	WINDOWS-AGENT	T1087 T1059.003	Discovery, Execution	Discovery activity spa...	3	92033
Jul 25, 2026 @ 11:09:06.905	WINDOWS-AGENT	T1087	Discovery	Discovery activity exe...	3	92031
Jul 25, 2026 @ 11:06:10.552	WINDOWS-AGENT	T1059.003	Execution	Powershell process s...	4	92004
Jul 25, 2026 @ 10:59:46.694	WINDOWS-AGENT	T1546.011	Privilege Escalation, Persistence	Application Compatibi...	12	92058
Jul 25, 2026 @ 10:45:41.519	WINDOWS-AGENT	T1087	Discovery	Discovery activity exe...	3	92031
Jul 25, 2026 @ 10:45:41.519	WINDOWS-AGENT	T1087	Discovery	Discovery activity exe...	3	92031
Jul 25, 2026 @ 10:45:23.682	WINDOWS-AGENT	T1059.003	Execution	Powershell process s...	4	92004
Jul 25, 2026 @ 10:40:14.886	WINDOWS-AGENT	T1059.003	Execution	Powershell process s...	4	92004
Jul 25, 2026 @ 10:37:26.346	WINDOWS-AGENT	T1087	Discovery	A net.exe account dis...	3	92039
Jul 25, 2026 @ 10:37:26.333	WINDOWS-AGENT	T1087	Discovery	A net.exe account dis...	3	92039
Jul 25, 2026 @ 10:37:26.321	WINDOWS-AGENT	T1087	Discovery	A net.exe account dis...	3	92039
Jul 25, 2026 @ 10:37:26.306	WINDOWS-AGENT	T1087	Discovery	A net.exe account dis...	3	92039
Jul 25, 2026 @ 10:37:25.295	WINDOWS-AGENT	T1087	Discovery	A net.exe account dis...	3	92039
Jul 25, 2026 @ 10:37:25.293	WINDOWS-AGENT	T1087	Discovery	A net.exe account dis...	3	92039

Results

Successfully collected:

- Process Creation
- Windows Security Events
- Sysmon Operational Logs
- PowerShell execution
- Command Prompt execution
- User information
- Parent-child process relationships

Skills Demonstrated

- SIEM deployment
- Windows endpoint monitoring
- Sysmon deployment
- Event collection
- Threat hunting
- Windows log analysis
- Process monitoring
- Endpoint visibility
- Security event investigation

Key Learning Outcomes

- Learned how Sysmon enhances Windows telemetry.
- Configured Wazuh to ingest Sysmon Operational logs.
- Investigated Windows process creation events using Wazuh dashboards.
- Understood the relationship between Windows Event Viewer, Sysmon, and SIEM event pipelines.
- Practiced troubleshooting agent connectivity and log collection issues.